



What participants learned from a standard operating procedure–based e-learning course in calf care: A mixed methods evaluation

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ABSTRACT

Training of new employees on dairy farms is often inadequate due to limited time, staff, expertise, and language barriers. This study evaluated the acceptance and learning outcomes of e-learning courses as a training tool for dairy farm workers. We developed an online course series based on standard operating procedures for 8 key tasks in calf care (neonatal care, emergency care, bottle feeding colostrum, tube feeding colostrum, measuring colostrum quality, disbudding, identification of sick calves, and cleaning and disinfection) and made it available to farmers and their employees (n = 301). Courses were published sequentially over a period of 3 mo. Five embedded questionnaires, including an open text field (“What was the most important thing you learned?”), were used to assess demographics, knowledge, and perceptions. A mixed methods approach combined quantitative analysis with qualitative content analysis of feedback. The main objectives of the study were to (1) explore dairy employees’ pretraining knowledge using qualitative approaches, (2) assess the effectiveness of the course in improving explicit and implicit knowledge of calf care, and (3) examine participants’ perceptions of e-learning as a format for continuing education. Results showed that employees were highly motivated to learn but often lacked essential basic knowledge in calf care such as time and volume of colostrum feeding or importance of general hygiene and measurement of colostrum quality. Self-perceived learning effects were high (49%–80% agreement), especially in the emergency care course. Interest was initially high but declined in later published courses. In terms of participation 24 h after publishing, emergency care was the most popular, whereas cleaning and disinfection was the least. In the final evaluation (n = 120), 90% rated the courses as easy to use and 97% as understandable. More than 85% ex-

pressed strong interest in continuing education via online courses, despite 68% (n = 301) never having taken one before. A chi-squared analysis revealed a significant association between future interest in e-learning and both education level and job position, with employees and higher-educated participants showing greater interest. These findings suggest that e-learning can effectively fill training gaps in dairy farming and support workforce development in the sector.

Key words: employee training, standard operating procedures, calf care, e-learning

INTRODUCTION

With increasing herd size, the success of the dairy farm depends more and more on a hired workforce (Durst et al., 2018). Often, dairy employees are immigrants, and language barriers occur (Schenker and Gunderson, 2013; Barkema et al., 2015; Durst et al., 2018). Furthermore, today, employees often do not have a dairy background (Hagevoort et al., 2013; Erskine et al., 2015; Alanis et al., 2022). Therefore, training is needed (Erskine et al., 2015) and gives farms an advantage in recruiting and retaining inexperienced employees (Durst et al., 2018; Moore et al., 2020). The latter is significant as high employee turnover causes intensive costs (Billikopf and González, 2012).

Generally, on-farm training is a rare event often conducted by coworkers or on the job (Erskine et al., 2015; Durst et al., 2018; Sischo et al., 2019; Alanis et al., 2022). Such an approach can cause protocol drift and a lack of basic knowledge (Román-Muñiz et al., 2007), which is critical because employees make important decisions about animal health and treatment (Román-Muñiz et al., 2007; Schuenemann et al., 2013; Erskine et al., 2015). Human resource management is one of the owners’ main challenges when expanding their dairy herds (Bewley et al., 2001; Hagevoort et al., 2013). However, owners underestimate the importance of a systematic approach to building a training culture (Alanis et al., 2022) and employees’ interest in learning (Durst et al., 2018).

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

One reason for inadequate on-farm training is that dairy producers often lack time or experience in adult education to train an inexperienced workforce (Maloney, 1992; Román-Muñiz et al., 2007; Erskine et al., 2015), especially when there is high employee turnover and language barriers exist. The need for and importance of alternative approaches to individual on-farm training are undisputed (Alanis et al., 2022; Neukirchner et al., 2024). Despite benefits such as social interaction with instructors and other participants and extended feedback, traditional training methods, such as centralized workshops and on-farm instruction, face important constraints. Centralized workshops are often impractical for dairy workers, who work long hours and may not be able to attend multihour sessions that require travel (Erskine et al., 2015). Similarly, although on-farm instruction is more accessible, the availability of experienced bilingual trainers is limited. E-learning courses are considered an alternative (Alanis et al., 2022) as they are available at any time and from any place. Compared with in-person training, there are additional benefits, including various formats, ease of translation, individual learning speeds, and repetition. In previous studies, we have shown that online training increases the feeling of confidence in work performance among dairy employees (Hesse et al., 2019; Neukirchner et al., 2024) and has the potential to improve milker behavior toward best-practice routines (Heuwieser et al., 2024).

The challenge of developing e-learning for a diverse audience is that the participant's knowledge and skill levels are unknown until feedback is obtained after the completion of a given course. Additionally, the extent of qualitative research focusing on dairy employees is low (Ritter et al., 2023).

Therefore, the overall objective of this study was to evaluate online courses based on standard operating procedures (SOP) for training dairy farm employees in relevant tasks related to calf care. Specifically, we aimed to (1) assess dairy employees' pretraining knowledge using qualitative methods, (2) evaluate whether such courses effectively convey information about calf care, capturing both explicit knowledge gains and implicit learning insights, and (3) determine if participants view e-learning positively for their continuing education.

MATERIALS AND METHODS

The Ethics Committee of Charité Berlin granted ethics approval for the study under application number EA1/254/19. We updated 3 previously used online courses (Hesse et al., 2019) and created 5 new ones providing best management practices for dairy farms on calf health, focusing on 8 topics: neonatal care, emergency care, bottle feeding colostrum, tube feeding colostrum, measur-

ing colostrum quality, disbudding, identification of sick calves, and cleaning and disinfection. The courses were designed using a cloud-based authoring software (Gomo Learning, Brighton, UK) and released in sequence from November 2019 to February 2020.

The findings were divided into 2 parts as the study generated abundant data. The results on the prevalence and usefulness of SOP on German dairy farms were published previously (Neukirchner et al., 2024), and the results about the perception of and knowledge gain through e-learning are presented in this paper.

Course Creation

Detailed information about the dissemination and the content of the courses and questionnaires has been described by Neukirchner et al. (2024). In brief, the courses were based on SOP, which provide step-by-step instructions. Each SOP included 3 sections: "What do I need?" "How do I do it?" and "Why is it important?" These sections are effective in employee training (Erskine et al., 2015; Hesse et al., 2019; Mills et al., 2020).

The first 2 sections were created as step-by-step guides, with each step featuring a brief, imperative title of 2 to 4 words and a description of 1 to 3 sentences about the task, accompanied by a clear image or video clip.

The third section, "Why is it important?" posed relevant questions related to the task, such as "Why is the first milk important?" or "Why is pain relief important during disbudding?" The answers were highlighted with a few sentences and intuitive figures or images to illustrate specific reasons for completing each step precisely as described in the SOP—a practice that has been shown to enhance learning motivation (Włodkowski, 2008).

Each course included a quiz, enabling participants to assess their understanding of the content. The courses were mobile-friendly and accessible to registered users through a URL. Participants could access the courses and individual sections at their convenience, revisiting them as often as needed throughout the study period. Each course was designed to be completed in about 8 min.

Questionnaires

Participants were asked to complete a total of 5 questionnaires (published in the Appendix of Neukirchner et al., 2024). At registration, we gathered information on demographics, prior training experience, and details about the farm (Q1). During the coursework we administered 3 questionnaires (Q2–Q4). The information obtained with Q2 and Q3 is not within the scope of this study. At the end of each course (Q4), a questionnaire was presented that contained 7 questions (5 Likert scale questions and 2 questions with 2 answer options) to evaluate the learning

effect and the learning environment. One open text field (“What was the most important thing you learned?”) was also provided. In the final evaluation (Q5), we asked about technical issues, appreciation of the courses, perception of online training post-training, and user engagement.

We employed multiple-choice questions and statements rated on 5-point Likert scales. As participants could pause the training and resume it later, the number of responses varied across questions in each course questionnaire. If a participant accessed a questionnaire multiple times, only their initial response to each question was included in the analysis.

Branding of Courses and Recruiting of Participants

The 8 courses were collectively branded under the name “Calf School” (Kälberschule) and were accessible in German through registration on a study website. The study was promoted from June to November 2019 at continuing education events for dairy farmers and their employees, as well as through dairy industry stakeholders, agricultural institutions, and farm magazines (via website posts, direct mailings, and social media). Individuals interested in participating registered on the study website, providing an email address for further communication. Participation was voluntary, with no incentives offered. Courses were provided free of charge, and participants were not required to complete them.

Study Procedure

The courses were released sequentially, starting with the first course published on November 28, 2019, and concluding with the final course on February 11, 2020. Each course remained accessible for the entire study period, which ended on February 28, 2020. Participants received an email invitation for each new course launch, followed by reminder emails sent 2 to 6 d later.

Statistical Analysis: Quantitative

A total of 440 individuals registered for the study, with 301 dairy farm employees (79.8%) included in the final dataset for analysis. The exclusion criteria were (multiple exclusion criteria applied in some cases): users not working on a farm (63), those who provided implausible information in the registration questionnaire (13) or an incorrect email address (6), individuals who registered after the course start of November 28, 2019 (56), or those under 18 yr of age (7) at registration. Due to the inclusion of partially completed questionnaires, sample sizes varied across questions.

A formal sample size calculation was not conducted because the study was exploratory and intended to iden-

tify typical engagement patterns within this target group. The number of participants ($n = 301$) was deemed more than enough to observe overall trends in participation and completion rates, based on previous similar studies on e-learning usage, where the participant count was significantly lower (Hesse et al., 2019; Alanis et al., 2022).

Two questions, “I have the following education . . .” and “I work on the farm as . . .,” were categorized and Likert scale responses were condensed from a 5-point scale to a 3-point scale (strongly agree or agree; neither agree nor disagree; disagree or strongly disagree) for analysis, except for the chi-squared test and contingency tables.

The course software tracked the launches per user and per course. When a user accessed a course multiple times, only data from the first launch were included and subsequent data were removed.

All data were initially collected in Microsoft Excel (Microsoft Office 2021, Microsoft Deutschland Ltd.), where descriptive statistics and graphs were also prepared. Data management and additional descriptive analyses, including means, standard deviations, frequency tables, contingency tables, and chi-squared tests were performed using SPSS Statistics for Windows (version 26.0, SPSS Inc., IBM). We also used R (version 4.2.1, R Core Team) and RStudio (version 2022.07.2, RStudio Team) to merge datasets and create initial Likert scale plots, which were finalized using GIMP (version 2.10.34) and Microsoft PowerPoint (Microsoft Office 2021, Microsoft Deutschland Ltd.).

Qualitative Analysis

For the open text field of Q4 (“What was the most important thing you learned?”), we employed qualitative content analysis as described in Mayring (2014) to systematically capture the emerging themes by course.

Separate tables were created for each course, listing the answers with one row for each pseudonymized participant. Responses were excluded if the entries contained only one digit (e.g., “.” or “1”) or one word (e.g., “very” or “calves”) to limit potential bias from interpreting overly brief answers. Obvious spelling and grammatical errors were corrected. A total of 60, 50, 39, 39, 33, 28, 27, and 16 coding units remained for the courses neonatal care, emergency care, bottle feeding, tube feeding, colostrum quality, disbudding, identification of sick calves, and cleaning and disinfection, respectively. Coding units varied in length from 2 to 92 words, with a median of 7 words written.

The documents were uploaded to an open-access web application (<https://www.qcmap.org/ui>) for systematic text analysis based on the techniques of qualitative content analysis (Mayring, 2022). The text was screened for codes and segments relevant to the research question. We

conducted an inductive category assignment to formulate categories while working through the text, iteratively revising the categories and rules for selection and category creation. We developed main categories where applicable.

Two steps were implemented to ensure the intersubjective objectivity and reliability of the analysis. After the first author (S.N.) coded the first course (neonatal care), the second author (K.C.J.) repeated the coding of this data subset to evaluate intercoder reliability. The results were compared and discrepancies were discussed. The first author then started coding all courses from scratch. When topics appeared in multiple courses, category names were standardized, and an intracoder agreement was performed using the final categories.

First, we addressed the actual research question (“What was the most important thing you learned?”) for each course individually. For each course, we counted the number of entries per category to determine both the total number and the relative frequency.

While reviewing the documents, patterns emerged in the text passages that did not align with the original research question. Therefore, we decided to re-analyze the documents with a second question in mind: “What was important for the participants to mention in several courses beyond the actual learning effect?”

Quotes were chosen to demonstrate examples of specific categories within each theme. Quotes have been edited for brevity and clarity, with ellipses indicating omitted text. Parentheses provide information about which course the statement was made in.

RESULTS AND DISCUSSION

In brief, the participants’ ($n = 301$) ages averaged 36.6 ± 11.6 yr (mean $\pm$ SD), ranging from 18 to 61 yr, with 5.4% of the participants being older than 55 yr. This percentage is much lower than the percentage reported for the whole population of German farmers with 37% being older than 55 (Deutscher Bauernverband e.V., 2022). Young people tend to be more interested in e-learning (Statista Consumer Insights, 2024), and younger farmers are more inclined toward technology overall (Vecchio et al., 2020; von Keyserlingk et al., 2024), which explains the relatively low age in our study. Of the participants, 111 (36.9%) were male, and 190 (63.1%) were female.

About two-thirds (207; 68.8%) held agricultural degrees, including apprenticeships and animal science degrees, whereas 13.6% (41) possessed other degrees or apprenticeships. The remaining 17.6% (53) had no further education beyond elementary or secondary school, or they lacked agricultural training. More demographic details are described elsewhere (Neukirchner et al., 2024).

Regarding roles, 31.9% (96) were owners or farm managers, and 68.1% (205) were farm employees. Con-

cerning specialization, 16.3% (49) worked with calves most of the time, 73.8% (222) worked with calves, but also in other areas, and 10.0% (30) did little or nothing with calves.

Farm sizes varied, with 22.6% (68), 57.5% (173), 12.6% (38), and 7.3% (22) working on farms with fewer than 100, 100 to 500, 500 to 1,000, and more than 1,000 cows in milk, respectively. Therefore, most participants (77.4%) were employed on farms with more than 100 cows in milk. This aligns well with national farm demographics, which indicate that about 75% of cows in Germany are kept on farms exceeding 100 cows (Deutscher Bauernverband e.V., 2022).

Evaluation of the Courses

The self-perceived learning effect achieved was high in most courses, particularly in the emergency care course (80% agreement; Figure 1). This may be because emergency care is well established in the veterinary domain, while many concepts were unfamiliar to the participating dairy employees (e.g., drugs for respiratory stimulation, resuscitation). In contrast, the learning effect for neonatal care was relatively low (49% agreement) compared with the other courses, likely because this is a prevalent task for the participants (separation of cow and calf, navel dipping, first colostrum feeding, documentation).

The self-perceived learning effect (49%–80% agreement) was high, despite most participants having an agricultural degree. Using the course classification of Neukirchner et al. (2024), tasks were grouped into frequently performed tasks (class 1: neonatal care, bottle feeding, identification of sick calves, and cleaning and disinfection) and rarely performed tasks (class 2: emergency care, tube feeding, colostrum quality assessment, and disbudding). The self-perceived learning effect was greater for class 2 than for class 1 tasks (Figure 1). We attribute this difference to participants’ lower self-perceived confidence before training and the marked increase in confidence from pre- to post-training in class 2, which amplified the perceived learning effect (Neukirchner et al., 2024).

To answer the second research question of the qualitative analysis (“What was important for the participants to mention across several courses beyond the actual learning effect?”), open-text responses ($n = 292$) were sorted into 3 categories: (1) feedback about the content, (2) feedback about the course design, and (3) other feedback.

Category 1: Feedback on the Content of the Courses

Totally New. Some participants indicated that the learning content was entirely new to them. This could be for several reasons. For some, the task was not part of

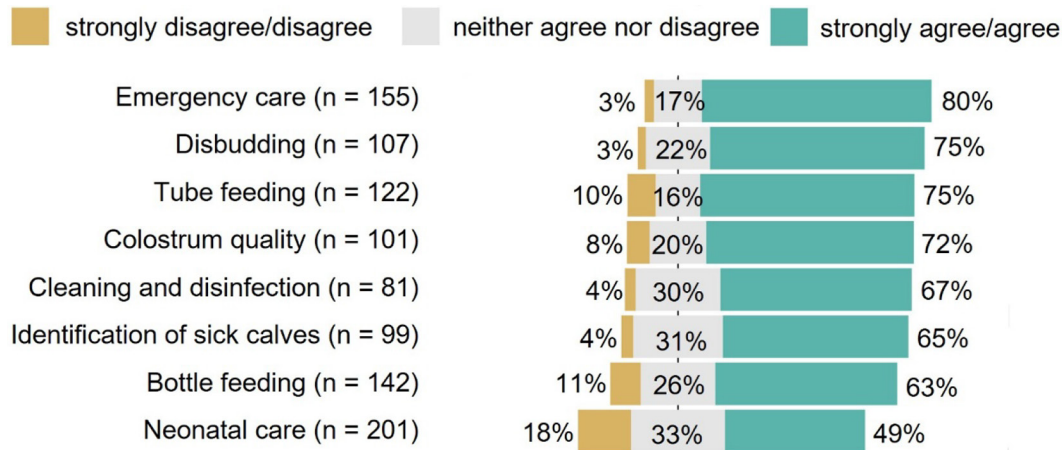


Figure 1. Agreement (%) with the statement “I’ve learned a lot from the course” on a 3-point Likert scale, presented as a stacked bar chart.

their daily job routine, as one participant described: “It was all completely new to me—I never disbud calves” (disbudding). The fact that this participant took the course on disbudding, despite not having to perform the task, shows a strong interest in continuing education in general. For other participants, some content was entirely new because what we considered a standard routine in calf care, such as measuring colostrum quality, was obviously not conducted on their farm. For one participant, it was even new “that there are measuring devices [for colostrum]” (colostrum quality).

Feedback in this category was only provided for rarely performed courses of class 2 (emergency care, disbudding, colostrum quality, and tube feeding) with 6, 5, 3, and 1 responses, respectively. These tasks were associated with a high increase in self-perceived confidence (Neukirchner et al., 2024) and learning effect after training.

Just Basics. Others stated that they were already familiar with the course content: “Nothing new for me, these are all basics” (neonatal care) or that they had different expectations about the course content: “I had hoped for and expected more detail and expert knowledge from the course. The course was very basic for me” (neonatal care). The fact that both extremes were covered in the comments illustrates how different the prior knowledge of the target group was. Creating online training materials for dairy employees in calf care is challenging because participants have widely varying levels of prior knowledge. Many calf caretakers did not grow up on farms or receive formal education in this area before starting their jobs (Erskine et al., 2015; Sisco et al., 2019), making it challenging to design content that engages everyone effectively. We assume that in the future, artificial intelligence (AI) can play an important role in customizing and personalizing learning content to better meet individual and farm-specific needs.

Theory Versus Practice. The size and structure of dairy farms in Germany vary greatly (Dachrodt et al., 2021), which can make some SOP presented in the training less suitable for certain farms. For example, one participant in the neonatal care course noted, “For us as a family business without employees, however, it is not very practical because it is too labor-intensive.” A similar concern was expressed in the course on bottle feeding colostrum: “[. . .] Because calves usually stay with their mothers for 8 to 10 hours. Then such a protocol is not possible, as no people are present 24/7.” Nevertheless, we consider the instruction to administer 4 L of colostrum within 4 h after birth to be critical. Although our previous work demonstrated that SOP were generally perceived as suitable across most farms (Neukirchner et al., 2024), these findings highlight the potential need for farm-specific adaptations. Furthermore, a shortage of skilled labor can affect proper SOP implementation: “As always, there are major differences between theory and practice: lack of personnel and qualifications [. . .]” (emergency care).

Other challenges can hinder the implementation of best practices, such as regulations and the availability of a veterinarian. Both were exemplified in the following comments in the course on disbudding: “Every federal state, in some cases every state veterinary office and every regulatory authority, has its own ideas on how to do it properly and the vet has no time, as he only injects xylazine himself” and “Unfortunately, the disbudding course was not very useful for me personally, as the procedure described does not fit into my routine (i.e., calling the vet every time).”

Study participants estimated the feasibility of the disbudding SOP to be the lowest. We presented a combination of possible pain control options, including a nonsteroidal anti-inflammatory drug, a sedative (xylazine), and a local anesthetic (cornual nerve block), as regulations

regarding disbudding have changed over recent years and are not consistent across all federal states of Germany. Moreover, federal laws in Germany mandate that local anesthesia can only be applied by veterinarians, which requires a veterinarian to be present for disbudding if a cornual nerve block is used, whereas the application of xylazine can be performed by the farmer if a veterinary-client-patient relationship exists.

Some farmers may lack the knowledge that pain must be treated or underestimate the importance of their personal responsibility for it (Sumner et al., 2018). Although online courses can address knowledge gaps, information on better management practices alone does not automatically cause behavior change (Ritter et al., 2017).

Category 2: Feedback About the Course Design

Visualization. A large number of participants expressed appreciation for the course design, particularly 3 aspects: structure (SOP)—“I would like your explanations to remain accessible for a long time. Everything is logically structured and presented in very good sections. Thank you for your work. It helps me” (identification of sick calves); footage—“The videos and pictures made the procedure really easy to understand” (disbudding); and downloadable material. In the neonatal care course, we provided a downloadable, predesigned infographic with fields for relevant calf information (e.g., calf ID, birth date, weight, volume of colostrum fed) to be used on calf hutches. It was well received, as evidenced by statements such as, “I will put an information board on each calf hutch.” In the course on identifying sick calves, the SOP outlined a daily examination routine and recommended using the Wisconsin Calf Health Scoring Chart (McGuirk and Peek, 2014) to assess sick calves. We provided a downloadable translated scoring chart referencing the original source. It was regarded as an added value to some of the participants. The rationale for this approach was to motivate and engage dairy employees using visual aids (Román-Muñoz et al., 2007) and translation of relevant findings into easy-to-understand text and video messages (Ritter et al., 2017).

Confidence. Previously, we showed that the SOP-based e-learning courses improved the perceived confidence in correctly performing certain calf care tasks (Neukirchner et al., 2024). The qualitative analysis of participants’ input in the open text field confirms this result. One participant wrote in the tube feeding course, “I was unsure about getting into the right tube. I was always worried about letting the milk run into the lungs. Now I know how to do it and how to control it.” The lack of confidence in performing the task before taking the course suggests that fewer calves were drenched by this employee, even though they might have benefited from

tube-fed colostrum. This demonstrates the potential of online training to improve calf health and animal welfare.

The videos were also mentioned again as a benefit: “I’ve never dehorned because I didn’t learn how, and I would never have dared to. The course was great, and the videos provide confidence” (disbudding).

It is noteworthy that the online training also built confidence to challenge common issues or failures: “I [will] try to convince my colleague to do it the same way” (neonatal care) and “Now I know exactly how to do it properly. Sometimes I’ve watched it and thought: ‘That can’t be right’” (disbudding). We infer that participants perceived the courses as credible because they were developed by veterinarians from the University of Berlin’s Veterinary College, a professional group widely trusted by dairy producers (Sumner et al., 2020). Receiving sound confirmation provides the confidence to stand up to your colleagues, as this participant in the neonatal care course realized: “That my running after them because of a lack of cleanliness is fully justified.”

In Neukirchner et al. (2024), 85% rated our courses as suitable for training new employees. Supporting comments include: “A very well explained instruction for employees!” (cleaning and disinfection) and “A very good basis for apprentices and career changers [. . .]” (emergency care). The latter suggests that even critical tasks such as emergency care are often performed by untrained staff.

Category 3: Other Feedback

Comparison with Peers. To illustrate task implementation in detail, we used video footage from large dairy farms. Participants valued these insights into practices on other farms as one noted in the neonatal care course: “My farm is already quite well prepared [. . .]. It is always interesting to see how different the requirements and initial conditions are on other farms.” Comparable reactions related to such comparisons with peers have been reported by Sumner et al. (2020), for example: “I like to listen to what other farms are doing [. . .].” Our step-by-step instructions further enabled participants to compare each procedure with their own practices, as reflected in one farmer’s comment during the bottle feeding course: “[I’ve learned] that things are different here than the latest studies suggest.” Benchmarking is recommended to enhance farmers’ problem awareness and sense of responsibility (Ritter et al., 2017) as well as their motivation to improve calf care (Atkinson et al., 2017; Sumner et al., 2020).

Several participants identified ways to improve their farm after taking our courses: “I have learned that we should change a few things on the farm” (identification of sick calves), “We do not have an emergency kit in our

operation. It was very informative. I will bring it up for our team” (emergency care). This stimulated discussion with their colleagues, supervisor, or with the veterinarian: “I realized we need to revise the procedure with our veterinarian” (disbudding).

Repetition. “It is important that we always control what we do. Nothing is worse than letting down a little creature that relies on us. Accordingly, repetition is also important (time well spent). So, this school is a good thing” (neonatal care). This participant expressed genuine empathy and responsibility for the calves, highlighting the value of repetition. Most participants agreed or strongly agreed (81%–92%, $n = 99$ –202) that the courses provided valuable repetition of knowledge, with only minor distinctions between the topics. One participant considered repetition to be essential for reducing farm blindness: “I have many years of experience with calving and calves, but repetition of how to do it right is very important. You must not become blind to the business” (neonatal care).

Awareness. Some farmers tend to assign less value to calves compared with cows, for example, by not calling a veterinarian for a single sick calf (Sumner et al., 2020). Participation in our courses increased awareness of animal welfare issues, as reflected in statements such as, “It does matter if one is ill” (identification of sick calves). The courses also addressed knowledge gaps: “There are many other things that I did not know before” (emergency care) and emphasized overlooked aspects of task performance: “I was made aware of the fact that many things are overlooked in stressful everyday life” (neonatal care). These findings underscore the importance of continuous training and regular re-evaluation of tasks and SOP.

What Did They Learn?

The results of the qualitative content analysis for the question, “What was the most important thing you learned?” are shown in Table 1. These insights help identify participants’ pretraining knowledge and can guide others in designing effective training programs for dairy employees by highlighting relevant and often-overlooked topics.

Despite decades of research aimed at guiding large dairy operations in calf rearing, critical gaps in routine best practices persist. Notable weaknesses include the volume, quality, and timing of colostrum feeding (Atkinson et al., 2017) and hygiene practices during calving and the neonatal period such as umbilical disinfection, sanitation of calf pens, and cleaning of feeding equipment (Hayer et al., 2021). Course feedback summarized in Table 1 shows that many employees lack fundamental knowledge rather than motivation. Farm consultancy and training programs should therefore prioritize the transfer of foundational concepts.

Additionally, concerns about animal welfare in society have grown, increasing pressure on farmers (Sumner et al., 2018). Participant feedback in the courses on emergency care, tube feeding colostrum, disbudding, and identification of sick calves shows a strong commitment to calf welfare (Table 1). Employees value training that helps them support animal well-being, making these topics likely to be well received. These themes identified in our study are consistent with findings from other qualitative research on calf rearing (Sumner et al., 2018; Sumner et al., 2020; Russell et al., 2022).

It is important to recognize the exploratory nature of this study and the limitations in interpreting category frequencies. The number of open-text responses was relatively small ($n = 16$ –60, see Table 1) compared with the total number of participants in the courses ($n = 105$ –247, see Table 2). This raises the possibility of response bias, where participants who learned something new may have been more likely to respond, whereas those already familiar with the material might not have responded.

The frequency with which a topic was mentioned does not necessarily reflect a widespread knowledge gap or its importance to most participants. Some topics, such as colostrum quality, may have been recalled more often simply because they were emphasized across multiple courses (e.g., neonatal care, bottle feeding, tube feeding, and colostrum quality). Because each course was designed as a stand-alone, repeating key concepts was intentional. Participants taking multiple courses may have shifted their attention to new content in later sessions.

Despite these limitations, our findings offer initial insights and highlight the importance of assessing pre-training knowledge among dairy employees. We encourage further research in this field to support the design of employee-centered training.

Perception of E-Learning

The average response rate (number of individuals who opened the course divided by the number of people invited) was 55.2% ($n = 301$), slightly higher than in a comparable study (48%; Hesse et al., 2019). Only 13.3% of registered participants did not access any course (Neukirchner et al., 2024). A total of 74 participants (24.6%) completed all 8 courses.

Participation declined almost linearly over the course series, from 247 participants (82.1%) in the first course on neonatal care to 105 (34.9%) in the final course on cleaning and disinfection (Table 2). This suggests that maintaining a high participant engagement over a prolonged training period is challenging. An exception was the course on colostrum quality, which had an unexpectedly low participation rate, possibly because it was released shortly before Christmas (December 17)

Table 1. Qualitative content analysis for the question “What was the most important thing you learned?” (categories with total count and frequency distribution; n = number of valid answers)

Category	No. of entries		Representative quote
	Absolute	Percent ²	
Neonatal care (n = 60)	10	16.7	“Gloves should be worn on newborn calves.”
Hygiene (gloves, boots, calf taxi, in general)	6	10.0	“Pay more attention to the cleanliness of the calf taxi.” “That a calf gets 4 liters of colostrum in the first 4 hours.” “4 liters in 4 hours (i.e. offering colostrum a second time if it only drank 3 liters the first time, for example).”
Colostrum: volume, time of feeding, multiple offerings	6	10.0	“That birth weight is important.” “That cold stress in calves starts at 10°C.” “Tasks must always be performed in the same way between individual employees and the necessary points must be documented.” “Mucus removal and stabilization for better breathing.”
Weighing	6	10.0	“Structured work can make many things easier and better if you do it.”
Cold weather stress and warming	5	8.3	“[...] navel disinfection with iodine. We use blue spray.”
Documentation	5	8.3	“Didn’t know much about emergency care. Won’t hang [the calf] anymore. Will get some medication. Never found out about a mucus extractor, will get one.” (Quote 1) “I really liked the instructions on ventilation. A good resource for my daily work.” See quote 1. “So far we have poured cold water on the chest, but this will not be done in future [...].” See quote 1. See quote 1. “[...] The heat supply will also have to be improved in the first few weeks.”
Breathing (mucus removal, chest-abdominal position)	4	6.7	“Recognizing a strong or weak calf was the most important thing for me, as you can then possibly plan the next steps.”
Calf taxi (usefulness, hygiene)	4	6.7	“We do not have an emergency kit in our company. It was very informative. I will bring it up at ours. The use of the tools was also new to me. Great!”
Soft skill: standardization is important	3	5.0	“Do not give up on weak calves too early.”
Navel disinfection (iodine, disposable cup)	2	3.3	“It was unknown to me what minimum Brix value should be used for colostrum.”
Soft skill: repetition is important	2	3.3	“That calves should drink 4 liters of colostrum within 4 hours.”
Colostrum: temperature while feeding	1	1.7	“2 hours after the first colostrum feeding, if necessary, top up to 4 liters.”
Colostrum: quality (Brix threshold value, measurement)	1	1.7	“Clean equipment to prevent the transmission of pathogens.”
Nest score	1	1.7	“That colostrum should not be pooled.”
Importance of neonatal care	1	1.7	“[...] Use a Brix refractometer to measure how many [antibodies] reach the calf.”
Emergency care (n = 50)	1	2.0	“Blood test was interesting.”
Hand-powered vacuum pump	16	32.0	“How important colostrum is.”
Cold water pouring (neck only)	6	12.0	
Respiratory stimulant drug (doxapram)	5	10.0	
Do not suspend calf by the hind legs	4	8.0	
Warming	4	8.0	
Sneezing reflex	4	8.0	
Drinking weakness	1	2.0	
Assessment of calves	1	2.0	
Assemble emergency kit	1	2.0	
Importance of warming calves	1	2.0	
Soft skill: perseverance pays off	1	2.0	
Bottle feeding (n = 39)	8	20.5	
Colostrum: quality (Brix refractometer)	5	12.8	
Colostrum: quantity, time of feeding, multiple offerings	4	10.3	
Hygiene	4	10.3	
Do not pool colostrum	4	10.3	
Use a bottle (without a handle for ease of cleaning)	3	7.7	
Blood test (lg supply)	3	7.7	
Importance of colostrum	3	7.7	
Colostrum: temperature while feeding and warming up	2	5.1	
Colostrum: time of milking	1	2.6	

Continued

Table 1 (Continued). Qualitative content analysis for the question “What was the most important thing you learned?” (categories with total count and frequency distribution; n = number of valid answers)¹

Category	No. of entries		Representative quote
	Absolute	Percent ²	
Tube feeding (n = 39)			
Position control (palpation on the neck)	11	28.2	“What I didn’t know was that you can feel the tube on the left side of the neck.”
Inserting the tube correctly	6	15.4	“I was unsure about getting into the right tube. I was always worried about letting the milk run into the lungs. Now I know how to do it and how to control it.”
Calf handling during drenching	4	10.3	
Bag drencher (existence, usefulness)	2	5.1	
Colostrum: temperature while feeding	1	2.6	“What it sounds like when the tube is wrong.”
Position control (listening)	1	2.6	
Colostrum: time of feeding	1	2.6	
Colostrum: time of milking	1	2.6	
Flow rate	1	2.6	
Colostrum quality (n = 33)			
Importance of colostrum quality and measurement (Ig)	5	15.2	“Everyone on the farm should carry out the colostrum test. This gives the newborn calf the best start in life.”
			“How important it is to measure the value in the colostrum.”
How the Brix refractometer works	5	15.2	
Refractometer as a method for measuring quality	5	15.2	“We check it by eye and flow, so it’s new territory.”
			“That there are measuring devices.”
			“I didn’t even know the function of the device. Never worked with it before.”
Brix threshold value (limit for good colostrum)	2	6.1	“The limit for the Brix value—when you start talking about usable colostrum—was again very helpful.”
			“Minimum value 22 Brix”
Brix refractometer maintenance	2	6.1	
Quality instead of quantity	1	3.0	
Hygiene	1	3.0	
Good colostrum also from heifers	1	3.0	
Disbudding (n = 28)			
Local anesthesia and cornual nerve block	5	17.9	“I didn’t know that you could anesthetize between the eye and the horn.”
Medication (which/time management: duration and onset of action)	3	10.7	“Time management is very important when disbudding.”
			“Meloxicam only takes effect after 20 min [. . .]”
Aspiration before injection	2	7.1	“[. . .] Aspirate because procaine must not enter the bloodstream.”
General anesthesia (systemic and local)	2	7.1	
Do not remove horn bud	2	7.1	
Legislation varies from state to state	1	3.6	
Hygiene	1	3.6	
Need to evaluate procedure with the veterinarian	1	3.6	
Identification of sick calves (n = 27)			
Breathing assessment (auscultation/breathing rate)	8	29.6	“Breaths per minute”
			“Listen to the lung with a stethoscope. Unfortunately, our vet doesn’t show me how to do this [. . .]”
Health checks take time	2	7.4	“Plan time and not just in between.”
Health check daily and preferably every morning	2	7.4	“Always check calves at the same time—preferably first thing in the morning.”
Order of the examination steps	2	7.4	
Calf health score	2	7.4	“The board with the assessment and scoring.”
Hygiene (boot disinfection and general)	2	7.4	“I didn’t realize how important it was to disinfect the boots. There is no way to disinfect boots on our farm.”
Importance of the health check	1	3.7	“That it does matter if one is ill.”

Continued

Table 1 (Continued). Qualitative content analysis for the question “What was the most important thing you learned?” (categories with total count and frequency distribution; n = number of valid answers)¹

Category	No. of entries		Representative quote
	Absolute	Percent ²	
Cleaning and disinfection (C/D; n = 16)			
Selection of the correct C/D agent (depending on temperature)	3	18.8	“The handling and selection of disinfectants (summer vs. winter).” “[...] Aldehydes are not effective below 10°C.”
How a disinfectant works	2	12.5	“How the disinfectant works.”
Disinfection of teat buckets	1	6.3	“Always disinfect as well. We don’t do this with the buckets.”
C/D order	1	6.3	“The order, choice of products, tips, and advice on avoiding mistakes were very helpful. There is a lot in the topic that can be done right or often wrong in daily work. [...]”
Contact plate test to check success	1	6.3	“Clean teat bucket properly.”
Cleaning of teat buckets	1	6.3	“The importance of cleaning and disinfection.”
Importance of C/D	1	6.3	“Calf hutches empty for a few days [...]”
Empty time for calf hutches	1	6.3	“[...] cleaning is more effective with foam [...]”
Cleaning with foam is more effective	1	6.3	

¹C/D = cleaning and disinfection.²Do not sum to 100% because of multiple or off-topic entries.

or because its relevance was perceived as low. A similar pattern of declining participation has been reported in both online (Hesse et al., 2019) and in-person studies (Vázquez et al., 2025) on dairy employee training.

When evaluating participant interest in specific topics based on access rates, it is important to consider that the courses were available for varying durations. However, this potential bias appears limited: 33.9% of participants accessed a course within the first 24 h of publication and 71.1% within the first week. Using first-day access as a measure for topic interest, emergency care emerged as the most frequently accessed topic, whereas cleaning and disinfection was the least popular. The latter may have been viewed as routine and was additionally influenced by being the final course in the series. The appeal of emergency care is likely related to perceived knowledge gaps, its emotional impact, and high practical relevance. This interpretation is supported by the highest self-reported learning gains and qualitative feedback such as, “A topic where you rarely find information or training. Nobody likes to talk about calves that die after birth. At least on our farm, we often have nicer topics and I never talk about this subject with colleagues.” Such responses suggest that e-learning can be an effective means of addressing sensitive yet crucial knowledge gaps.

Despite some decline, participation remained relatively high. Each course attracted at least one-third of the participants, and 1 in 4 remained engaged through all 8 courses over a 3-mo period.

Data on e-learning use in this target group are limited, making interpretation difficult. It is likely that the email was the main factor driving engagement as participants were notified via email with the course URL included, and most accesses occurred within 1 wk, shortly afterward. The average email open rate for consumer marketing in German-speaking countries in 2019 was 27.5% (Inxmail, 2024). Even the least accessed course (34.9%) exceeded this, indicating a high level of interest. In the final evaluation (n = 120), most participants rated the courses as easy to use (90%) and understandable (97%). Importantly, 85% expressed interest in using e-learning for future continuing education, even though 206 of the original 301 participants (68.4%) had never taken an online course before.

A chi-squared analysis revealed a statistically significant association between participants’ interest in future e-learning and both education level [$\chi^2(3) = 9.16$, $P = 0.027$, Cramér’s $V = 0.276$] and job role [$\chi^2(3) = 9.63$, $P = 0.022$, Cramér’s $V = 0.283$], each indicating a medium effect size (see Table 3). Participants with higher education levels were more likely to express interest in continuing education online, aligning with trends in the general population (Statista Consumer Insights, 2024), which may reflect greater familiarity with digital tools. In contrast, participants with lower education levels

Table 2. Number of participants and days online by course; the courses are listed in order of publication

Course	Participation		Days online	Participation 24 h after publishing		Participation 1 wk after publishing	
	Absolute	Percent ¹		Absolute	Percent ²	Absolute	Percent ²
Neonatal care	247	82.1	94	77	31.1	205	83.0
Emergency care	193	64.1	89	81	42.0	152	78.8
Bottle feeding	167	55.5	86	58	34.7	130	77.8
Tube feeding	158	52.5	82	57	36.1	98	62.0
Colostrum quality	119	39.5	75	42	35.3	79	66.4
Disbudding	151	50.2	47	48	31.8	94	62.3
Identification of sick calves	128	42.5	34	42	33.6	86	68.8
Cleaning and disinfection	105	34.9	19	28	26.7	73	69.5

¹The percentage refers to the total number of participants accessing at least 1 course (n = 301) = response rate.

²The percentage refers to the total number of participants per course.

did not show a pronounced preference for this format. Despite these group differences, overall interest in e-learning was high across all education levels and job roles. Regarding job role, dairy employees expressed greater interest in online continuing education than farm managers or owners. This preference may stem from the flexibility of e-learning, as in-person events often occur in the evenings and require travel, which is challenging after a full workday (Erskine et al., 2015). Furthermore, most in-person continuing education opportunities are designed for managers, limiting access for employees and increasing their interest in alternative, accessible formats like e-learning.

The detail in responses to “What was the most important thing you learned?” showed strong engagement with the courses and underscored the importance of culturally and linguistically tailored training (Ritter et al., 2017). Online formats offer an advantage through easy translation and customization for diverse dairy teams (Durst et al., 2018; Neukirchner et al., 2024).

Study Limitations

This study used a convenience sample with voluntary participation, which limits the generalizability of findings related to job roles, education, and interest in online

continuing education. Participants may have already had a positive attitude toward online training. Greater participation from people without agricultural education would have provided additional valuable insights. Because questionnaires were embedded in the course and required responses to continue, some participants might have given meaningless answers, introducing selection bias. To reduce this bias and gain more detailed insights, other more in-depth methods such as in-person interviews could have been helpful and might have provided additional insights. However, the advantage of the method used was not only its feasibility but also the anonymity of an online survey. This likely reduced the potential for social desirability bias.

We cannot confirm data saturation because all data were collected during a single period and analyzed retrospectively. Some responses were unique, whereas others were shared by about 30% of participants.

The study was conducted just before the COVID-19 pandemic was declared (March 11, 2020; World Health Organization, 2020). Since then, online course participation (Eurostat, 2022) and the use of learning management systems have increased (Moodle, 2024). Although a repeat study might produce different results today, the prepandemic timing offers an unbiased view of initial attitudes toward online continuing education.

Table 3. Agreement in percent with the statement “Online training is a good way for me to continue my education” after training considering education level and job role (n = 120)

Agreement	Education level		Job role	
	Agricultural or other degree/apprenticeship (n = 95)	No further education (n = 25)	Farm owner or manager (n = 30)	Employee (n = 90)
Strongly agree	72	48	30	61
Agree	16	36	47	27
Neither agree nor disagree	8	16	23	11
Disagree	4	—	—	1
Strongly disagree	—	—	—	—

CONCLUSIONS

Dairy farm employees value continuing education and see online courses as a useful tool. Beyond new knowledge, dairy farm employees value the chance to refresh skills, boost confidence through best-practice comparisons, and learn about other farm operations. Employers should support this interest by directing staff to credible, well-designed online training, especially when in-person options are limited. We speculate that employees prefer personalized e-learning that builds on their existing knowledge and appreciate features like videos and downloadable materials.

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Nonstandard abbreviations used: AI = artificial intelligence; C/D = cleaning and disinfection; Q1–Q5 = questionnaires 1–5; SOP = standard operating procedures.

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